

Unit 7 – Inference for Quantitative Data: Means

Notes 1: Confidence Intervals for Means

Recall: Sampling Distribution of μ

Shape	Center	Spread
Condition: _____, the _____ tells us it will be approximately normal If the population is normally distributed, we can reasonably assume the sampling distribution will be approximately normal.	Condition: The sample must have either: • been _____ from the population • used _____ in an experiment	Condition: When sampling, the 10% condition must be met:

Recall: Confidence Intervals

Why do we have confidence intervals? _____

Confidence Interval: _____

Confidence Interval for a Mean:

Where z^* is the critical value with _____ of the area lying between $-z^*$ and z^*

Example: Constructing a Confidence Interval for a Mean (With a Known σ)

A farmer wants to estimate the mean weight of all tomatoes grown on his farm. To do so, he selected a random sample of 4 tomatoes, and calculated the mean weight to be 110g. Assuming the population of all the tomatoes in his field are normally distributed, and that the standard deviation of his tomatoes in the past has been 40g, use the sample mean to create a 99% confidence interval for the population mean.		
<i>Step 1</i>	<i>Communicate:</i> Communicate the parameter in the context of the problem and write down the confidence level you are using	

Step 2	<i>Conditions:</i> Check the conditions required (random, independent, normal)	
Step 3	<i>Calculate:</i> Identify the inference method and construct the confidence interval by writing down the formula, the values substituted into the formula, and then the interval itself.	
Step 4	<i>Conclude:</i> Interpret the interval in the context of the problem using the script.	

When we constructed this confidence interval, we made the assumption that we knew the true population standard deviation, σ . *This, of course, is a silly assumption!* If we are creating a confidence interval to give us an interval of plausible values for the population mean, how can we possibly know the population standard deviation (which we needed the population mean to determine!).

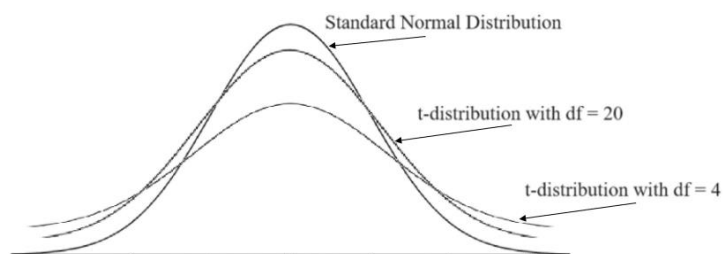
When we do not know the population standard deviation, we must estimate it from our sample using the sample standard deviation, s . However, when we do so, the test statistic, z , that we previously used, **changes**. The new test statistic is now called the t-statistic, and has a new distribution associated with it.

The Student's t Distribution

- The t-distribution (also called Student's t Distribution) is a family of related distributions which depend on a parameter called degrees of freedom.
- Degrees of freedom is tough to understand, but we use it to identify which t-distribution we are working with. It is denoted as _____ and it is easy to calculate for a t-distribution: _____
- t-distributions are unimodal, symmetric, and bell-shaped, just like a Normal distribution, but t-distributions with only a few degrees of freedom have longer tails and a larger standard deviation than the Normal.

The tails on a t-distribution are “fatter” than a standard normal distribution. This is true because the smaller our sample size, the more variation we have, hence the fatter tails.

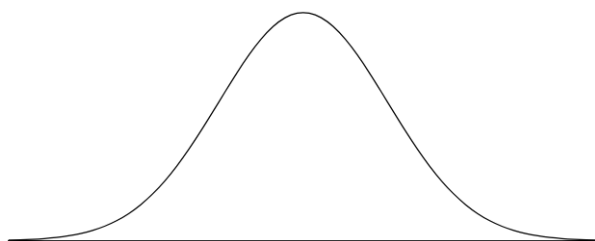
The larger our sample size gets (and larger dfs we get) the closer the t-distribution will move towards the standard normal distribution.



Calculating t^*

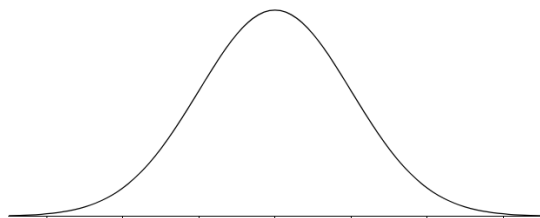
On the TI-84: Go to DISTR and down to 4: invT

The area listed has to be the tail probability of the C% interval you are finding:

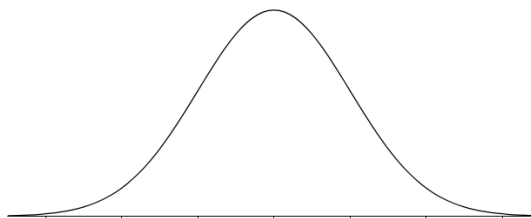


Find t^* for the following situations:

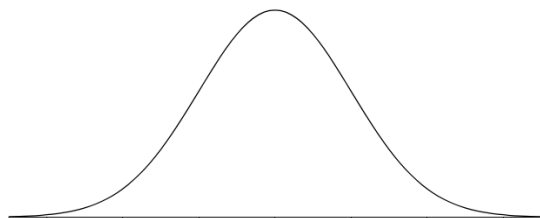
90% Confidence Interval with $n = 5$



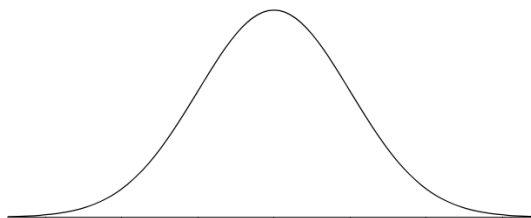
90% Confidence Interval with $n = 25$



95% Confidence Interval with $n = 18$



99% Confidence Interval with $n = 36$



Calculating a Confidence Interval for a Mean (Unknown σ)

When conditions are met, a C% confidence interval for the unknown proportion μ is:

where t^* is the critical value for the t-distribution with C% of its area between $-t^*$ and t^*

This is called a _____

<i>Random</i>	<i>Independent</i>	<i>Normal</i>
OR	When sampling, check the 10% condition; In an experiment, we do not have to check this	_____, the _____ tells us it will be approximately normal _____, we have to either: 1) Have the original population be approx normally distributed 2) Graph the sample data to make sure there are no outliers or strong skewness

Example: You recently read an article that said averaging 300 minutes of exercise a week will help fend off cancer. A random sample of 10 US adults was taken and they were asked how many of minutes of exercise they get each week. The data is given below.

265	274	281	289	304	243	275	290	325	286
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Construct a 95% confidence interval to estimate the true population mean. Based on this confidence interval, do you believe that adults are exercising enough to ward off cancer? Explain.